

Matrix Operations Project

Overview

This assignment implements various matrix operations in C++ using two-dimensional arrays and dynamic memory allocation. The program demonstrates fundamental concepts of matrix manipulation that are widely used in mathematics, image processing, and artificial intelligence.

Features

The project supports the following operations:

- Display a matrix
- Calculate the sum of matrix elements
- Compute the transpose of a matrix
- Input a convolution kernel
- Apply a filter (convolution) with zero padding
- Print a matrix in spiral order
- Check whether a matrix is upper triangular
- Check whether a matrix is lower triangular
- Check whether a matrix is an identity matrix
- Reverse a matrix
- Rotate an image (matrix rotation)
- Multiply two matrices

Functions

```
void display(int Matrix[][5], int m, int n);
int sum(int Matrix[][5], int m, int n);
void transpose(int Matrix[][5], int m, int n);
void input_kernel(int Matrix[][5], int** Matrix1, int a, int b, int m, int n);
void spiral_print(int Matrix[][5], int m, int n);
void Apply_filter(int Matrix[][5], int** Matrix1, int a, int b, int m, int n);
bool upper_triangular(int Matrix[][5], int m, int n);
bool lower_triangular(int Matrix[][5], int m, int n);
bool identity_matrix(int Matrix[][5], int m, int n);
void Reverse(int Matrix[][5], int m, int n);
void Rotate_image(int Matrix[][5], int m, int n);
void product(int** Matrix, int** Matrix1, int m, int n, int m1, int n1);
```

AI Relevance

Some of the implemented operations are directly related to Artificial Intelligence and Neural Networks:

- **Matrix Multiplication:** Used in neural network computations.
- **Transpose:** Used during backpropagation.
- **Apply Filter:** Similar to convolution operations in Convolutional Neural Networks (CNNs).
- **Kernel Input:** Represents filters used for feature extraction.
- **Image Rotation:** Can be used for data augmentation.

Technologies Used

- C++
- Dynamic Memory Allocation
- Two-Dimensional Arrays
- Functions and Modular Programming